

A RANDOMIZED CONTROLLED TRIAL

COMPARING HIGH-DOSE VERSUS LOW-DOSE WEIGHT ADJUSTED STEROID REGIMEN IN THE TREATMENT OF

IDIOPATHIC GRANULOMATOUS MASTITIS



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BACKGROUND

Corticosteroids remain the first line preferable breast-conserving oral option; however, prescribing practice remains highly heterogeneous, with no standardized dosing protocol, leading to variable resolution rates and unpredictable steroid-related morbidity. The trial is designed to compare the rate of complete clinical resolution (CCR) of IGM at ten weeks between high-dose and low-dose regimen, while evaluating which regimen provides a more favorable safety profile.

OBJECTIVES

Primary objective is to determine the most effective weight-adjusted corticosteroid regimen for achieving complete disease resolution in patients with idiopathic granulomatous mastitis. Secondary objectives are association between clinical and radiological resolution and rate of side effects.

METHODS

- A multi-center single-blinded RCT across 5 centers in Malaysia
- 36 patients with HPE confirmed IGM recruited over 2 years (1 August 2024 to 31 July 2026)
- 1:1 randomization to high (1.0 mg/kg/day) or low-dose (0.5 mg/kg/day) weight-adjusted prednisolone arms, both tapered fortnightly over 10 weeks
- Breast ultrasound at baseline and 10 weeks.
- Intention-to-treat analysis using SPSS v25 (Shapiro-Wilk normality testing; independent t-test/ Mann-Whitney U for continuous variables; Fisher's exact/Freeman-Halton exact test for categorical variables).

RESULTS

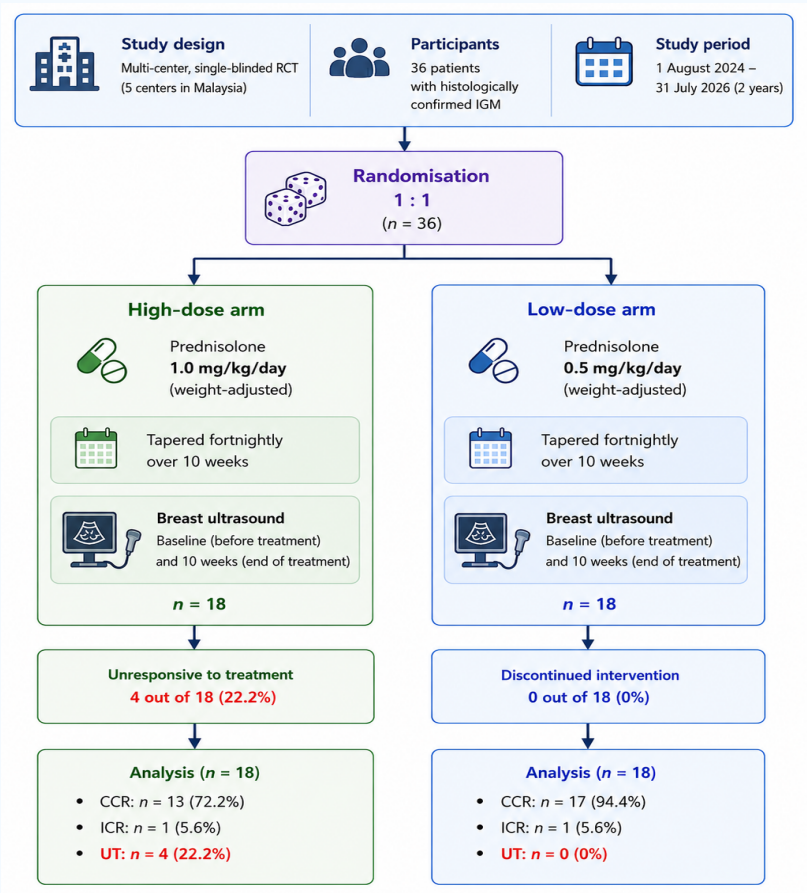
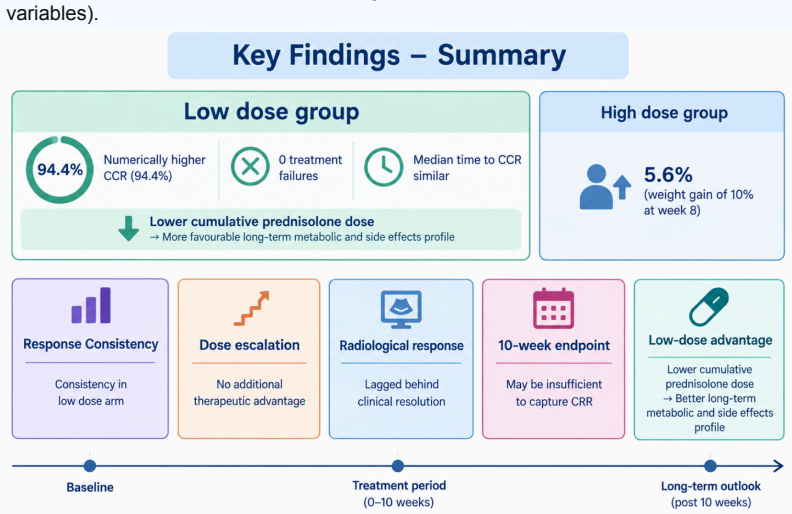
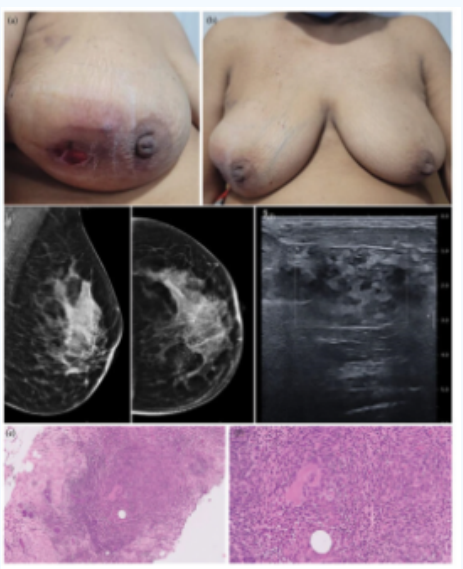


FIGURE 1





CONCLUSION / TAKE-HOME MESSAGE

1. Main finding

- Low-dose weight-adjusted prednisolone (0.5 mg/kg/day) achieved a numerically higher complete clinical resolution rate at 10 weeks (94.4% vs 72.2%, p=0.072)
- Zero treatment failures versus 22.2% in the high-dose arm
- Fewer side effects (0% vs 5.6%)

2. Clinical meaning

- Findings do not support dose escalation as a strategy to improve outcomes in IGM
- Low-dose prednisolone appears to be the appropriate starting regimen, offering comparable clinical response with a lower cumulative steroid burden and more favourable safety profile

3. Next step

- Double-blind, placebo-controlled design
- Extended follow up
- Integration of validated QOL & cosmetic outcomes

REFERENCES

1. Bede K, Valente SA. Idiopathic granulomatous mastitis. Annals of Breast Surgery. 2020 Dec;4:24–24.
2. Kessler E, Wolloch Y. Granulomatous Mastitis: A Lesion Clinically Simulating Carcinoma. Am J Clin Pathol. 1972 Dec 1;58(6):642–6.
3. Schelfout K, Tjalma WA, Cooremans ID, Coeman DC, Colpaert CG, Buytaert PM. Observations of an idiopathic granulomatous mastitis. Eur J Obstet Gynecol Reprod Biol. 2001 Aug;97(2):260–2.
4. Tang A, Dominguez DA, Edquilang JK, Green AJ, Khoury AL, Godfrey RS. Granulomatous Mastitis: Comparison of Novel Treatment of Steroid Injection and Current Management. Journal of Surgical Research. 2020 Oct;254:300–5.
5. DeHertogh DA, Rossof AH, Harris AA, Economou SG. Prednisone management of granulomatous mastitis. N Engl J Med. 1980 Oct 2;303(14):799–800.